

MTL106: Probability and Stochastic Processes

(Lecture 2)

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Axiomatic Approach of Probability: In the modern (axiomatic) approach of probability theory, we do not concerned about how probabilities are assigned to some events in the sample space Ω . We define the probability for certain types of subsets of the sample space Ω . For this, we use some axioms that are consistent with the three properties of the classical as well as relative frequency approaches. The axiomatic probability approach is based on some concepts of the set theory and measure theory. We first define these concepts and then move to the axiomatic definition of probability.

Class of Sets: A set whose elements are themselves set is known as a class of sets. Eg., $\mathcal{A} = \{\{1\}, \{1, 2\}, \{2, 3, 4\}\}$ is a class of sets.

Set Function: Let $\mathcal{C}$ be a class of sets. A function $\mu : \mathcal{C} \rightarrow \mathbb{R}$ is known as a *set function*. This means a set function is a real valued function whose domain is a class of sets.

Remark 1: Axiomatic approach deals with the probability assignments to the members of an appropriately chosen class $\mathcal{C}$ of subsets of the sample space Ω . Eg., (1) if $\Omega = \mathbb{R}$, then $\mathcal{C}$ may be a class of all open intervals in $\mathbb{R}$, (2) If Ω is a countable set, then $\mathcal{C}$ may be a class of all singletons $\{\omega\}$, $\omega \in \Omega$.

Remark 2: The members of the class of sets $\mathcal{C}$ are known as *basic sets*.

Event Space and Events: Suppose $\mathcal{F}$ is a class of sets for which the probability assignment to be done. Then the class $\mathcal{F}$ is known as the *event space* and elements of $\mathcal{F}$ are called *events*. Also $\mathcal{F}$ satisfies the following properties:

1. $\Omega \in \mathcal{F}$,
2. $E \in \mathcal{F} \Rightarrow E^C \in \mathcal{F}$,
3. $E_j \in \mathcal{F} \Rightarrow \bigcup_{j=1}^{\infty} E_j \in \mathcal{F}$, $j = 1, 2, \dots$

σ -Field: A σ -field of subsets of sample space Ω is a class $\mathcal{F}$ of subsets of Ω which satisfies the following properties:

1. $\Omega \in \mathcal{F}$,
2. $E \in \mathcal{F} \Rightarrow E^C \in \mathcal{F}$,

$$3. E_j \in \mathcal{F} \Rightarrow \bigcup_{j=1}^{\infty} E_j \in \mathcal{F}, j = 1, 2, \dots$$

It is clear that a σ -field is a non-empty class of subsets of Ω that is closed under the formation of countable unions and complements and contains the null set ϕ .

Remark 3: The event space is basically a σ -field.

Remark 4: $\mathcal{F} = \{\phi, \Omega\}$ is known as a *trivial* σ -field.

Remark 5: $\mathcal{F} = \{E, E^C, \phi, \Omega\}$ is the *smallest* σ -field containing the set E .

Remark 6: The couplet $(\Omega, \mathcal{F})$ is known as a *measurable space*.

Example 1: Let $\Omega = \{A, B, C\}$. Then one possible σ -field is

$$\mathcal{F} = \{\phi, \{A\}, \{B, C\}, \{A, B, C\}\}.$$

Example 2: It might be possible that a sample space can be a set of real numbers in some situations where outcomes are continuous and can take any value within a certain range. Eg., measurements of temperature, height, weight, distance, blood pressure, sugar level, air quality index, etc. are continuous and take values in an interval. If $\Omega = \mathbb{R}$, then $\mathcal{F}$ might be $\phi, \mathbb{R}$, any countable union of open intervals within $\mathbb{R}$, any countable intersection of open intervals within $\mathbb{R}$, complements of above sets within $\mathbb{R}$, etc.

Probability Function: Let $\mathcal{F}$ be a σ -field of subsets of Ω . A *probability function* (or a *probability measure*) is a set function $P : \mathcal{F} \rightarrow \mathbb{R}$ and satisfies the following three axioms:

1. $P(E) \geq 0 \quad \forall \quad E \in \mathcal{F}$,
2. $P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$, where $E_1, E_2, \dots$ is a countably infinite collection of mutually exclusive events ($E_i \cap E_j = \phi, i \neq j, i, j = 1, 2, \dots$),
3. $P(\Omega) = 1$.

Remark 5: The triplet $(\Omega, \mathcal{F}, P)$ is known as a probability space.